Candidate resolutions for interval and absolute-value problems

Seven proposed resolutions in OpenProblemsInNLA

This reader contains six self-contained manuscripts addressing seven entries in the repository section on intervals and absolute value equations. The arguments are presented as claimed resolutions for independent review. They have not been accepted by the repository, independently peer reviewed, or externally submitted by this work.

Claim index

AV-01 Exactly 2^n solutions of $Ax+|x|=b$ can be recognized with $n+1$ rational LP feasibility tests, without a regularity promise.

AV-02 The spectral inverse-norm threshold is NP-hard under a many-one reduction, even for integer upper-triangular matrices with diagonal 2. The decision problem on the corresponding rational class is NP-complete.

IV-02 Exact tridiagonal interval determinant ranges are NP-hard to compute, even for regular interval families. The upper determinant threshold is NP-complete.

IV-03 An interval matrix is inverse-M throughout if and only if n^2 specified vertices are inverse-M. This implies the proposed $2n^2$ criterion.

IV-04 Exact tridiagonal interval solution hulls are NP-hard to compute, even with a regular interval family and point right-hand side $-e_n$.

IV-05 An exact solution hull under the inverse-M interval promise is computable with $2n$ rational linear programs.

IV-06 A 3×3 interval matrix with integer endpoints has at least four connected components in its real eigenvalue set.

AV-03 and IV-01 are not claimed solved. The tridiagonal hardness result is a bit-complexity result and is not claimed to be strong NP-hardness or an approximation lower bound. The general exact-output questions for IV-02 and IV-04 are equivalent to $\mathsf{P}=\mathsf{NP}$, as explained in their joint manuscript.

Reviewing the package

Start with IV-06 for a short, exact counterexample. Its certificate consists of four integer eigenpairs and three excluded real numbers. AV-01 gives the shortest new structural characterization. One rational-rotation reduction then supplies both IV-02 and IV-04. The inverse-M recognition and hull arguments in IV-03 and IV-05 are independent except that the former can check the promise needed by the latter.

Use the PDF bookmarks to move between manuscripts; each manuscript retains its own pagination and bibliography. The accompanying archive contains LaTeX sources, seven issue drafts, exact verification scripts, diagnostic numerical scripts, and retained result files. The verification summary distinguishes completed passes from missing or incomplete runs. Finite computation supplements the proofs; it does not establish their general correctness.

The manuscripts are AI-generated research drafts. No human authorship or institutional affiliation has been supplied. Independent proof review, attribution decisions, and a further novelty check should precede public submission.

Recognizing the maximum finite number of solutions of an absolute value equation

Repository AV-01. Claimed resolution; independent review pending.

Abstract

For rational $A \in \mathbb{R}^{n\times n}$ and $b \in \mathbb{R}^n$, we give a polynomial-time test for whether $Ax+|x| = b$ has exactly 2^n distinct solutions. The test uses $n+1$ rational linear-programming feasibility problems. No nonsingularity or spectral-radius promise is required; infinite solution sets are rejected. The proof characterizes the maximum finite solution count by the geometry of a polytope with paired inequalities.

1 Statement and relation to the problem

Absolute values and inequalities are componentwise. Put $D_d = \operatorname{diag}(d)$ and

$$P = \{y : (A+I)y \le b,\ (A-I)y \le b\}, \qquad u(y) = b - Ay.$$

Thus $y \in P$ if and only if $u(y) \ge |y|$. The repository asks for polynomial-time recognition of exactly 2^n solutions, including the possibility of singular systems and infinitely many solutions in negative instances [1]. Earlier sufficient assumptions include a spectral-radius restriction [2].

Theorem 1. *The following are equivalent:*

1. *$Ax + |x| = b$ has exactly 2^n distinct solutions.*
2. *$b > 0$, P is bounded, and $u_i(y) > 0$ for every $y \in P$ and $i = 1, \ldots, n$.*

2 Necessity

In the closed orthant indexed by $s \in \{-1, 1\}^n$, the solution set is

$$\{x : (A + D_s)x = b,\ D_s x \ge 0\}.$$

It is convex. A finite such set contains at most one point. Count incidences between solutions and closed orthants. If there are exactly 2^n solutions, each orthant contains exactly one solution, and each solution belongs to exactly one closed orthant. In particular all its coordinates are nonzero. Denote the solution by $x_s = D_s u_s$, where $u_s > 0$. The matrix $A + D_s$ is nonsingular: otherwise its nullspace would give a nonconstant line segment of solutions in the open orthant. Consequently $N_s = I + AD_s$ is nonsingular and $N_s u_s = b$.

For $d \in [-1, 1]^n$, let

$$q(d) = \det(I + AD_d),$$

and let $r_i(d)$ be the determinant obtained by replacing column i of $I + AD_d$ by b. Both polynomials are multiaffine, and r_i does not depend on d_i. Cramer's rule gives

$$r_i(s) = q(s)(u_s)_i.$$

Flipping s_i leaves the left side unchanged, so the determinants at two adjacent cube vertices have the same sign. All cube vertices are connected by such flips. Their determinant average is $q(0) = 1$, hence every $q(s)$ is positive. Every $r_i(s)$ is therefore positive as well. Multiaffine interpolation yields

$$q(d) > 0, \qquad r_i(d) > 0 \quad (d \in [-1, 1]^n).$$

At $d = 0$ this implies $b_i = r_i(0) > 0$. The continuous function

$$u(d) = (I + AD_d)^{-1}b$$

is strictly positive on the entire cube.

For any $y \in P$, define $u = b - Ay \geq |y|$. Set $d_i = y_i/u_i$ when $u_i > 0$, and set $d_i = 0$ when $u_i = 0$ (which also forces $y_i = 0$). Then $y = D_d u$ and $(I + AD_d)u = b$, so $u = u(d) > 0$. Conversely, $y = D_d u(d)$ is in P for every d in the cube. Thus

$$P = \{D_d u(d) : d \in [-1, 1]^n\}.$$

This is the continuous image of a compact set, and all the asserted slack inequalities hold.

3 Sufficiency

Assume condition 2. The origin satisfies all $2n$ inequalities strictly, so P is nonempty and full dimensional. In pair i, the inequalities are $y_i \leq u_i(y)$ and $-y_i \leq u_i(y)$. They cannot both be active at a feasible point, since that would force $y_i = u_i(y) = 0$.

A vertex of a full-dimensional polytope requires n linearly independent active normals. At most one normal per pair can be active. Therefore each vertex has exactly one active inequality from each pair, and those n normals are independent. Its coordinates are nonzero, and it is an AVE solution.

We show that the sign patterns of vertices are closed under flipping any one coordinate. Start at a vertex and retain its $n - 1$ active equations from all pairs except pair i. Their common nullspace is one dimensional. Choose the direction that strictly relaxes the omitted active inequality. A sufficiently small step in this direction is feasible. Boundedness gives a finite last feasible point on the resulting ray. At this endpoint some new inequality must become active. It cannot be the opposite of any retained inequality, since two members of a pair cannot be active together. It cannot be the inequality being strictly relaxed. It must therefore be the opposite member of pair i. Its normal is independent of the retained normals, since its value changes along the ray. Hence the endpoint is a vertex with exactly sign i flipped.

A nonempty bounded polytope has a vertex. Repeated flips now produce a vertex in every open orthant. For each pattern, its active system is nonsingular, so there is at most one such vertex. Conversely, an AVE solution lies in P, is strict in sign, and satisfies one of these nonsingular active systems. It equals the corresponding vertex. There are exactly 2^n solutions, proving Theorem 1.

4 The algorithm and bit complexity

First reject if any $b_i \leq 0$. Test feasibility of

$$(A + I)r \leq 0, \qquad (A - I)r \leq 0, \qquad -\mathbf{1}^T Ar = 1. \tag{1}$$

Reject if feasible. Indeed, a recession vector satisfies $-Ar \geq |r|$. For every nonzero such vector, $-\mathbf{1}^T Ar \geq \|r\|_1 > 0$, so it can be scaled to satisfy the last equality. Conversely, any feasible vector in (1) is a nonzero recession vector. Since P is already nonempty, this is exactly the boundedness test.

For each i, test feasibility of

$$(A + I)y \leq b, \qquad (A - I)y \leq b, \qquad A_{i,:}y = b_i. \tag{2}$$

Reject if any of these n systems is feasible, and accept otherwise. On P, the final equality is precisely $u_i(y) = 0$; all u_i are nonnegative there. Theorem 1 proves correctness.

There are $n+1$ feasibility tests, each with n variables, $2n$ inequalities, and one equality. Their rational coefficients have polynomial encoding length in the input. Polynomial-time rational linear programming [3] therefore gives a polynomial bit-complexity algorithm. This statement concerns exact rational linear programming; floating-point implementations in the accompanying tests are only diagnostic.

Remark 2. Boundedness is essential. For $n = 1$, $A = 2$, $b = 1$, one has $P = (-\infty, 1/3]$ and $1 - 2y > 0$ throughout P, but the equation has only one solution.

5 An exact example beyond the spectral-radius restriction

For

$$A = \frac{3}{5}\begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}, \qquad b = (1,2)^T,$$

the four solutions are

$$(10/73, 95/73)^T, \quad (-10/7, 5/7)^T, \quad (40/7, -95/7)^T, \quad (-40/13, -5/13)^T.$$

Direct substitution verifies each equation, and the four sign systems are nonsingular. Meanwhile $|A|$ has spectral radius $6/5 > 1$. This example illustrates that the theorem is not confined to the earlier spectral-radius subclass.

References

[1] OpenProblemsInNLA, problem AV-01. `https://github.com/ajt60gaibb/OpenProblemsInNLA/tree/main/intervals-and-absolute-value-equations/AV-01`.

[2] M. Hladík, *Absolute value equations with 2^n solutions*, Optimization Letters 20 (2026), 559–575. `https://doi.org/10.1007/s11590-025-02251-z`.

[3] N. Karmarkar, *A new polynomial-time algorithm for linear programming*, Combinatorica 4 (1984), 373–395. `https://doi.org/10.1007/BF02579150`.

Hardness of a spectral-norm condition number for absolute value equations

Repository AV-02. Claimed resolution; independent review pending.

Abstract

We prove NP-hardness, under a polynomial-time many-one reduction, of deciding whether the maximum spectral norm of $(A - \operatorname{diag} d)^{-1}$ over $d \in [-1,1]^n$ is at least a rational threshold. Hardness holds for integer upper-triangular A with every diagonal entry equal to 2; the entire diagonal perturbation family is automatically regular. The decision problem restricted to rational upper-triangular matrices with this diagonal is NP-complete.

1 Decision problem

For a matrix whose diagonal perturbation family is regular, define

$$c_2(A) = \max_{d \in [-1,1]^n} \|(A - D_d)^{-1}\|_2.$$

The repository asks whether deciding $c_2(A) \geq t$ is NP-hard in the spectral norm [1, 2]. We use positive rational t; nonpositive thresholds are trivial.

Lemma 1. *For every regular diagonal perturbation family, the maximum defining $c_2(A)$ is attained at a vertex $d \in \{-1,1\}^n$.*

Proof. Fix all coordinates except $d_i = t$, and write the matrix at $t = 0$ as M. Sherman–Morrison gives

$$(M - te_ie_i^T)^{-1} = M^{-1} + \frac{t}{1-\alpha t}uv^T, \quad \alpha = e_i^T M^{-1} e_i, \quad u = M^{-1}e_i, \quad v^T = e_i^T M^{-1}.$$

The denominator never vanishes on $[-1,1]$. The scalar function $t/(1-\alpha t)$ has derivative $1/(1-\alpha t)^2 > 0$. Thus every inverse on this segment is a convex combination of the two endpoint inverses. Convexity of a norm implies that one endpoint has at least as large a norm. Move to an endpoint one coordinate at a time. Notice that convexity in t itself is not being asserted. $\square$

Theorem 2. *The decision problem $c_2(A) \geq t$ is NP-complete on rational upper-triangular matrices with diagonal entries all equal to 2. NP-hardness persists when A and t are integers of polynomial magnitude in the source graph size.*

2 Reduction from MAX-CUT

Use the standard NP-complete decision version of unweighted MAX-CUT [3]. Let G have v vertices and $m \geq 1$ edges, and let $1 \leq k \leq m$. Let $B \in \{-1,0,1\}^{v \times m}$ be an oriented incidence matrix. Set

$$N = 1 + v + m, \qquad K = 12N^2, \qquad A = \begin{pmatrix} 2 & K\mathbf{1}^T & 0 \\ 0 & 2I_v & KB \\ 0 & 0 & 2I_m \end{pmatrix}.$$

All perturbed diagonal entries lie in $[1,3]$. Every member of the family is therefore nonsingular.

Let a, the diagonal entries of X, and the diagonal entries of Y denote the reciprocals of the perturbed diagonal entries in the three blocks. They range independently over $[1/3, 1]$. Direct block inversion gives

$$(A - D_d)^{-1} = \begin{pmatrix} a & -Ka\mathbf{1}^T X & K^2 a\mathbf{1}^T XBY \\ 0 & X & -KXBY \\ 0 & 0 & Y \end{pmatrix}.$$

If C denotes the maximum cut size, then

$$\max_{x\in[1/3,1]^v} \|x^T B\|_2 = \frac{2}{3}\sqrt{C}. \tag{1}$$

Indeed, the squared norm is $\sum_{\{p,q\}\in E}(x_p - x_q)^2$. It is convex in each coordinate, hence has a maximizing box vertex. Each crossing edge then contributes $4/9$, and every other edge contributes zero.

The largest possible norm of the upper-right block is consequently $\frac{2}{3}K^2\sqrt{C}$, attained with $a = 1$, $Y = I$, and a maximum-cut vertex x. The spectral norm of the whole inverse is at least the norm of this subblock. The remaining blocks together have norm at most

$$1 + K\sqrt{v} + K\sqrt{2m} \le 3KN = 36N^3,$$

using $\|B\|_2 \le \|B\|_F = \sqrt{2m}$. Hence

$$96N^4\sqrt{C} \ \le\ c_2(A) \ \le\ 96N^4\sqrt{C} + 36N^3. \tag{2}$$

Compute the integer

$$q = \left\lfloor \sqrt{144N^2 k} \right\rfloor, \qquad t = 8N^3 q.$$

Integer square root is computable in polynomial time. If $C \ge k$, the lower bound in (2) gives $c_2(A) \ge t$. If $C \le k - 1$, then

$$\sqrt{k} - \sqrt{k-1} = \frac{1}{\sqrt{k} + \sqrt{k-1}} \ge \frac{1}{2N}$$

and $q > 12N\sqrt{k} - 1$. Therefore

$$t - 96N^4\sqrt{k-1} > 48N^3 - 8N^3 = 40N^3 > 36N^3.$$

The upper bound in (2) now gives $c_2(A) < t$. This is a polynomial-time many-one reduction. In particular it is stronger than Turing hardness. All entries of A and the threshold have polynomial magnitude in N.

3 Membership in NP

For the specified upper-triangular class, regularity requires no certificate. By Lemma 1, a yes certificate is a sign vector s. Write $M = A - D_s$. For $t > 0$,

$$\|M^{-1}\|_2 \ge t \quad \Longleftrightarrow \quad t^2 M^T M - I \text{ is not positive definite.}$$

The matrix on the right is rational. Sylvester's criterion tests positive definiteness by exact leading principal determinants in polynomial bit complexity. A zero determinant is correctly accepted as "not positive definite." Thus the certificate can be checked without approximate eigenvalue calculations. This proves Theorem 2.

Remark 3. No assertion is made here that the regularity promise can be recognized in NP for arbitrary input matrices. The NP-completeness statement is for the explicit upper-triangular diagonal-2 class, on which the promise is automatic. NP-hardness for the original promised problem follows immediately.

References

[1] OpenProblemsInNLA, problem AV-02. https://github.com/ajt60gaibb/OpenProblemsInNLA/tree/main/intervals-and-absolute-value-equations/AV-02.

[2] M. Zamani and M. Hladík, *Error bounds and a condition number for the absolute value equations*, Mathematical Programming 198 (2023), 85–113. https://doi.org/10.1007/s10107-021-01756-6.

[3] *MaxCut on Permutation Graphs is NP-complete*, primary research manuscript (2022), Theorem 1. https://arxiv.org/abs/2202.13955. The stronger restricted-graph result in particular implies the standard unweighted MAX-CUT hardness used here.

Exact tridiagonal interval determinants and solution hulls are NP-hard

Repository IV-02 and IV-04. Claimed resolutions; independent review pending.

Abstract

A reduction from PARTITION encodes signed sums of small rational rotation angles in the continuant of a tridiagonal interval matrix. It proves NP-completeness of the upper determinant threshold problem, and NP-hardness of exact determinant-range and exact solution-hull computation. The reduction uses independent interval entries, fixed superdiagonal entries equal to 1, and an everywhere nonsingular interval family. For the hull result the right-hand side is the single point $-e_n$. General exact-output upper bounds are also recorded, so each requested polynomial-time existence question is equivalent to $\mathsf{P} = \mathsf{NP}$.

1 Scope

The exact determinant-range and solution-hull problems for rational tridiagonal interval matrices are the subjects of IV-02 and IV-04 [1, 2]. Polynomial algorithms for important special subclasses do not resolve the general questions [3, 4, 5]. The reduction below does not assert strong NP-hardness: the rational gap can be small in the numerical value of the input.

2 Rational rotations as continuants

For rational a, β, put

$$M(a, \beta) = \begin{pmatrix} a & -\beta \\ 1 & 0 \end{pmatrix}.$$

Given a PARTITION instance with positive integer weights $w_1, \ldots, w_m$, let

$$W = \sum_i w_i, \quad \delta = \frac{1}{10W^2}, \quad t_i = \delta w_i, \quad c_i = \frac{1 - t_i^2}{1 + t_i^2}, \quad s_i = \frac{2t_i}{1 + t_i^2}.$$

Here $c_i > 0$, $c_i^2 + s_i^2 = 1$, and

$$R_i = \begin{pmatrix} c_i & -s_i \\ s_i & c_i \end{pmatrix} = M(-s_i/c_i, -1/c_i)M(s_i, -c_i)$$

is a rotation through $\theta_i = 2\arctan(t_i)$. Also,

$$M(0, -1)M(0, q) = \operatorname{diag}(1, -q), \qquad q \in [-1, 1].$$

Thus a pair of companion matrices implements a rotation, and another pair implements an independent reflection-or-identity gate at its endpoints.

Construct $N = 4m - 2$ layers, in order of application, as follows:

j	a_j	β_j	range
$4i - 3$	s_i	$-c_i$	$i = 1, \ldots, m$
$4i - 2$	$-s_i/c_i$	$-1/c_i$	$i = 1, \ldots, m$
$4i - 1$	0	q_i	$i = 1, \ldots, m - 1, \quad q_i \in [-1, 1]$
$4i$	0	-1	$i = 1, \ldots, m - 1.$

Let $T(q)$ have diagonal a_j, all superdiagonal entries equal to 1, and subdiagonal entry $T_{j,j-1} = \beta_j$ for $j \geq 2$. The unused β_1 is only a notational convenience in the transfer identity. The $m-1$ intervals $q_i \in [-1, 1]$ are distinct matrix entries and are mutually independent.

The continuant recurrence for leading determinants is

$$p_{-1} = 0, \quad p_0 = 1, \quad p_j = a_j p_{j-1} - \beta_j p_{j-2}.$$

Therefore $\det T = p_N$ is the first coordinate of

$$M(a_N, \beta_N) \cdots M(a_1, \beta_1)(1, 0)^T.$$

At a gate vertex, put $\epsilon_i = -q_i \in \{-1, 1\}$. The successive vector angles satisfy

$$\phi_1 = \theta_1, \qquad \phi_{i+1} = \theta_{i+1} + \epsilon_i \phi_i.$$

Consequently the endpoint determinants are exactly

$$\cos\Big(\sum_{i=1}^{m} \sigma_i \theta_i\Big), \qquad \sigma_i \in \{-1, 1\}, \quad \sigma_m = 1. \tag{1}$$

Every such pattern occurs: take $\epsilon_i = \sigma_i \sigma_{i+1}$. Fixing the last sign loses no zero signed sum, since a simultaneous sign reversal preserves zero.

3 Separation and regularity

The elementary integral bound

$$0 \leq 2t - 2\arctan t \leq \frac{2}{3}t^3 \qquad (t \geq 0)$$

gives

$$E := \sum_i |\theta_i - 2\delta w_i| \leq \frac{2}{3}\delta^3 W^3 = \frac{\delta}{150W} \leq \frac{\delta}{10}.$$

Also $\sum_i \theta_i \leq 2\delta W = 1/(5W) \leq 1/5$. Every determinant in (1) is therefore positive. A determinant is multiaffine in independent entries; throughout a box it is a convex combination of its vertex values. Thus every matrix in the entire interval family is nonsingular, not just its vertices. The determinant maximum is attained at a vertex.

If PARTITION has a solution, choose its signs with $\sigma_m = 1$. Then the angle in (1) has absolute value at most $\delta/10$, giving

$$\max_q \det T(q) \geq 1 - \frac{\delta^2}{200}.$$

If there is no partition, each integer signed weight sum has absolute value at least 1. Every endpoint angle then has absolute value at least $2\delta - E \geq 19\delta/10$, and at most $1/5$. For $|x| \leq 1$, Taylor's inequality gives $\cos x \leq 1 - x^2/3$. Hence

$$\max_q \det T(q) \leq 1 - \frac{361}{300}\delta^2 < 1 - \delta^2.$$

The rational threshold

$$\tau = 1 - \frac{\delta^2}{2}$$

strictly separates the two cases.

All numbers constructed above have $O(\log W)$ bits per entry. For example c_i and s_i use integers of sizes polynomial in W, not an exponentially long list of repeated rotations. There are $O(m)$ rows and interval entries. The reduction is polynomial in the binary input length.

Theorem 1. *Deciding whether the maximum determinant of a rational tridiagonal interval matrix is at least a rational threshold is NP-complete. NP-hardness holds even when every matrix in the interval family is nonsingular and every superdiagonal entry is fixed at 1. In particular exact determinant-range computation is NP-hard.*

Proof. Hardness is the reduction above from the standard NP-complete PARTITION problem [6]. Membership in NP follows from multiaffinity: a vertex is a polynomial-size certificate, and its rational determinant is computable exactly in polynomial time. □

4 Transferring hardness to a solution hull

For a tridiagonal $N \times N$ matrix with all superdiagonal entries 1, the corner cofactor identity is

$$(T^{-1})_{1N} = \frac{(-1)^{N+1}}{\det T}.$$

Indeed, deleting row N and column 1 leaves a triangular matrix with diagonal entries all 1. Our $N = 4m - 2$ is even and all determinants are positive. Take the point right-hand side $b = -e_N$. Then

$$x_1 = (T^{-1}b)_1 = \frac{1}{\det T} > 0.$$

The lower endpoint of coordinate 1 of the exact interval solution hull is therefore

$$\underline{x}_1 = \frac{1}{\max_q \det T(q)}.$$

It decides the determinant threshold by comparison with $1/\tau$.

Theorem 2. *Exact solution-hull computation for rational tridiagonal interval systems is NP-hard even for regular interval matrices, point right-hand side $-e_N$, and fixed superdiagonal entries all equal to 1. These instances have nonempty bounded solution sets.*

5 Exact-output upper bounds

For completeness, both general exact-output problems are in $\mathsf{FP}^{\mathsf{NP}}$. This also makes precise the "unless $\mathsf{P} = \mathsf{NP}$" qualification.

For determinants, multiply all vertex entries by a common denominator D whose bit length is polynomial in the input. Let H bound the absolute values of the resulting integers. The scaled determinant is an integer of absolute value at most $n!H^n$. Binary search with the NP vertex-threshold oracle finds its exact maximum and minimum in polynomially many queries; divide by D^n.

For hulls, write the interval data as $[C - R, C + R]$ and $[b_c - b_r, b_c + b_r]$. Rowwise independent interval evaluation gives the exact solution set

$$\Sigma = \{x : |Cx - b_c| \le R|x| + b_r\} = \bigcup_{s \in \{-1,1\}^n} P_s,$$

where

$$P_s = \{x : D_s x \ge 0,\ (C - RD_s)x \le b_c + b_r,\ (-C - RD_s)x \le -b_c + b_r\}.$$

Each P_s is a rational pointed polyhedron, since it is contained in an orthant. Nonempty such polyhedra have vertices, and finite coordinate optima are attained at vertices. After a common denominator is cleared in the coefficients and right-hand sides, let H bound their integer sizes and put $Q = n!H^n$, increasing H to at least 1. Cramer's rule bounds every vertex coordinate numerator in absolute value by Q and its denominator by Q.

Nonemptiness is in NP: guess s and a polynomial-bit feasible vertex. Unboundedness of coordinate i is in NP: guess a feasible sign polyhedron and a rational recession vector normalized to have ith coordinate 1 (or -1). Such certificates have polynomial encoding length by standard rational-polyhedron bounds. Because the union has finitely many members, an unbounded coordinate in the union is unbounded in at least one member.

When an endpoint is finite, the query $\exists x \in \Sigma : x_i \geq t$ is in NP by the same sign-polyhedron certificate argument, including the extra rational inequality. Binary search brackets the endpoint in an interval of width less than $1/(2Q^2)$ inside $[-Q, Q]$. At most one rational with denominator at most Q is in that bracket. Continued-fraction rational reconstruction recovers it in polynomial time. Lower endpoints are handled by $-x_i$. Empty sets and infinite endpoints are covered by the earlier oracle queries.

Thus a polynomial-time algorithm exists for either complete exact-output task if $\mathsf{P} = \mathsf{NP}$. The hardness theorems prove the converse. These upper bounds are included as standard complexity completion, not as an independent novelty claim.

References

[1] OpenProblemsInNLA, problem IV-02. `https://github.com/ajt60gaibb/OpenProblemsInNLA/tree/main/intervals-and-absolute-value-equations/IV-02`.

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An n^2-vertex characterization of inverse-M interval matrices

Repository IV-03. Claimed resolution; independent review pending.

Abstract

An interval matrix consists entirely of inverse-M matrices if and only if n^2 specified vertices are inverse-M matrices. This proves the proposed $2n^2$-vertex criterion and shows that half of those tests are unnecessary. The proof combines induction on principal submatrices, monotonicity of two-index Schur complements, and an adjugate lemma that rules out singularity without assuming regularity in advance. Zero entries, zero interval widths, and reducible matrices are allowed.

1 Definitions and main theorem

A nonsingular M-matrix is a real matrix B with nonpositive off-diagonal entries and $B^{-1} \geq 0$. A matrix A is inverse-M if it is nonsingular, $A \geq 0$, and A^{-1} has nonpositive off-diagonal entries. These definitions include reducible matrices.

Let $\boldsymbol{A} = [L, U] = [C - R, C + R]$, where $R \geq 0$. For $i = 1, \ldots, n$, let $z^{(i)}$ have entry -1 at position i and entry 1 elsewhere, and put $D_i = \operatorname{diag}(z^{(i)})$. Define

$$V_{ij} = C - D_i R D_j \qquad (1 \leq i, j \leq n).$$

The repository's proposed polynomial criterion uses both signs of this expression [1, 2]; an exponential vertex characterization is known [3].

Theorem 1. *Every matrix in $\boldsymbol{A}$ is inverse-M if and only if every one of the n^2 matrices V_{ij} is inverse-M.*

The reverse implication is the substantive one. Since these n^2 vertices are among the proposed $2n^2$ tests, the theorem implies that proposed characterization as well.

2 Basic closure facts

Lemma 2. *Every principal submatrix of a nonsingular M-matrix is a nonsingular M-matrix, and all its principal minors are positive. Every principal submatrix of an inverse-M matrix is inverse-M and has positive determinant. Every principal Schur complement of an inverse-M matrix is inverse-M, in particular entrywise nonnegative.*

Proof. Let B be a nonsingular M-matrix. Set $w = B^{-1}\mathbf{1} > 0$; strict positivity follows because a nonnegative invertible matrix has no zero row. The matrix $H = B \operatorname{diag}(w)$ has nonpositive off-diagonal entries and satisfies $H\mathbf{1} = \mathbf{1}$. Thus H and all its principal submatrices are strictly row diagonally dominant with positive diagonal. Write any such submatrix as $D(I - K)$ with positive diagonal D, $K \geq 0$, and $\|K\|_\infty < 1$. The Neumann series gives a nonnegative inverse. The determinant is positive because $\det(I - tK)$ is continuous and nonzero for $0 \leq t \leq 1$, and is 1 at $t = 0$. Positive diagonal scaling transfers these conclusions to the principal submatrices of B.

Now let $A = B^{-1}$ and partition the indices as S, T. Jacobi's complementary principal-minor identity gives

$$\det A_{SS} = \frac{\det B_{TT}}{\det B} > 0,$$

with the empty determinant interpreted as 1. Moreover,

$$A_{SS}^{-1} = B_{SS} - B_{ST}B_{TT}^{-1}B_{TS}$$

has nonpositive off-diagonal entries. Since $A_{SS} \geq 0$, it is inverse-M. Finally,

$$A_{TT} - A_{TS}A_{SS}^{-1}A_{ST} = (B_{TT})^{-1}$$

is inverse-M and nonnegative. □

Lemma 3 (Adjugate completion)**.** *Let $n \geq 3$. If all nonempty proper principal minors of A are positive and all off-diagonal entries of* $\operatorname{adj} A$ *are nonpositive, then* $\det A > 0$.

Proof. Suppose first $\det A = 0$. A nonzero principal minor of order $n-1$ implies $\operatorname{rank} A = n-1$. Thus $H = \operatorname{adj} A$ has rank one and strictly positive diagonal. Rank one implies

$$H_{12}H_{23}H_{31} = H_{11}H_{22}H_{33} > 0,$$

contradicting nonpositivity of the three off-diagonal factors.

Suppose instead $\det A < 0$, and put $B = A^{-1}$. Its diagonal is negative and its off-diagonal entries are nonnegative. Jacobi's identity implies that every nonempty principal minor of B is negative, since its complementary principal minor in A is positive, or equals 1 when the complement is empty. Choose any three indices and write the corresponding principal submatrix as

$$\begin{pmatrix} -a & p & q \\ r & -b & s \\ t & u & -c \end{pmatrix}, \qquad a,b,c > 0, \quad p,q,r,s,t,u \geq 0.$$

Its negative principal minors of order two imply $pr > ab$, $qt > ac$, and $su > bc$. Its determinant is

$$-abc + asu + bqt + cpr + pst + qru > 2abc > 0,$$

a contradiction. Both alternatives are impossible. □

3 Proof of the vertex characterization

Assume all V_{ij} are inverse-M. Each lower endpoint is nonnegative, since $(V_{ij})_{ij} = L_{ij}$. Thus $L \geq 0$ and every A in the interval is nonnegative.

We induct on n. For $n = 1$, the tested matrix is L, and positivity of L suffices. For $n = 2$, the vertex

$$V_{11} = \begin{pmatrix} L_{11} & U_{12} \\ U_{21} & L_{22} \end{pmatrix}$$

has positive determinant. Hence every interval member has

$$\det A = a_{11}a_{22} - a_{12}a_{21} \geq L_{11}L_{22} - U_{12}U_{21} > 0.$$

Its inverse has nonpositive off-diagonal entries, proving the assertion.

Let $n \geq 3$. Every proper principal interval inherits the vertex hypothesis: the local test for indices i, j is the corresponding principal submatrix of the global V_{ij}. Lemma 2 and induction imply that every proper principal submatrix of every $A \in \boldsymbol{A}$ is inverse-M. In particular all proper principal minors of every interval member are positive. This is an inductive proof, not an instruction to perform exponentially many algorithmic tests.

Fix distinct i, j, and put $S = \{1, \ldots, n\} \setminus \{i, j\}$. Define

$$f_{ij}(A) = a_{ij} - A_{iS}A_{SS}^{-1}A_{Sj}, \qquad u = A_{iS}A_{SS}^{-1}, \quad v = A_{SS}^{-1}A_{Sj}.$$

Here $u \geq 0$ and $v \geq 0$. For example, in the proper principal inverse-M block on $\{i\} \cup S$, the off-diagonal block of its inverse is $-\alpha^{-1}u$, where the scalar Schur complement $\alpha > 0$ by the ratio of positive principal determinants. Its nonpositivity proves $u \geq 0$. The argument for v uses the block on $S \cup \{j\}$.

Throughout the box, ordinary matrix differentiation gives

$$\frac{\partial f_{ij}}{\partial a_{ij}} = 1, \qquad \frac{\partial f_{ij}}{\partial a_{ik}} = -v_k, \qquad \frac{\partial f_{ij}}{\partial a_{kj}} = -u_k, \qquad \frac{\partial f_{ij}}{\partial a_{k\ell}} = u_k v_\ell \quad (k, \ell \in S).$$

All other entries are irrelevant to f_{ij}. These signs show that its minimum is attained with a_{ij} and A_{SS} at their lower endpoints, and A_{iS} and A_{Sj} at their upper endpoints. These are exactly their values in V_{ij}. Consequently

$$f_{ij}(A) \geq f_{ij}(V_{ij}) \geq 0.$$

The last inequality holds because $f_{ij}(V_{ij})$ is an off-diagonal entry of the two-index principal Schur complement of an inverse-M matrix; Lemma 2 makes that complement nonnegative.

The adjugate identity

$$(\operatorname{adj} A)_{ij} = -\det(A_{SS})f_{ij}(A)$$

now proves that every off-diagonal adjugate entry is nonpositive. This identity does not require A itself to be invertible: it follows by eliminating the invertible block A_{SS} and taking the appropriate cofactor of the resulting two-by-two block, or by polynomial continuity. Lemma 3 yields $\det A > 0$. Hence A^{-1} has nonpositive off-diagonal entries. Since $A \geq 0$, it is inverse-M. The forward implication of Theorem 1 is immediate because the test matrices belong to the interval.

4 Complexity and boundary cases

Form n^2 rational vertices. For each, check nonsingularity and entrywise nonnegativity, compute its inverse exactly, and check its off-diagonal signs. This takes $O(n^5)$ arithmetic operations using standard cubic inversion, with polynomial rational bit complexity. The arithmetic-operation count is not a claim of strongly polynomial bit complexity.

No strict positivity of off-diagonal entries or interval widths was assumed. The proof explicitly treats possible singularity of an interior full matrix through Lemma 3, rather than presuming regularity to justify inversion. Reducible and point intervals are included.

References

[1] OpenProblemsInNLA, problem IV-03. `https://github.com/ajt60gaibb/OpenProblemsInNLA/tree/main/intervals-and-absolute-value-equations/IV-03`.

[2] M. Hladík, *An overview of polynomially computable characteristics of special interval matrices*, Springer (2020), 295–310, Section 9. `https://doi.org/10.1007/978-3-030-31041-7_16`. Author version: `https://arxiv.org/abs/1711.08732`.

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Exact inverse-M interval solution hulls by $2n$ linear programs

Repository IV-05. Claimed resolution; independent review pending.

Abstract

For a rational interval matrix whose every member is inverse-M, and an arbitrary rational interval right-hand side, we compute the exact solution hull using $2n$ linear programs with n variables and $2n$ inequalities each. A complementary-parts linear program solves the relevant absolute value equations. A sign argument using the inverse-M promise proves that their solutions are attained coordinatewise hull endpoints. All claims are in exact rational bit complexity; floating-point test code is supplementary only.

1 Problem and notation

Write

$$\boldsymbol{A} = [C - R, C + R], \quad R \geq 0, \qquad \boldsymbol{b} = [b_c - b_r, b_c + b_r], \quad b_r \geq 0.$$

Assume every $A \in \boldsymbol{A}$ is inverse-M: $A \geq 0$, A is nonsingular, and A^{-1} has nonpositive off-diagonal entries. Its inverse is then a nonsingular M-matrix and has positive diagonal entries. The requested hull is the smallest coordinate box containing

$$\Sigma = \{x : Ax = b \text{ for some } A \in \boldsymbol{A}, b \in \boldsymbol{b}\}.$$

This is the problem in IV-05 [1, 2]. The promise suffices for the algorithm; it can separately be checked by the criterion proposed in the accompanying IV-03 manuscript.

2 A bijective absolute value map

For a fixed sign vector $\sigma \in \{-1, 1\}^n$, define

$$H_\sigma(x) = Cx - D_\sigma R|x|.$$

Every selection matrix $C - D_\sigma R D_s$, and indeed every matrix $C - D_\sigma R D_d$ for $d \in [-1, 1]^n$, belongs to $\boldsymbol{A}$.

Lemma 1. *If every matrix in $\boldsymbol{A}$ is nonsingular, then H_σ is a bijection of $\mathbb{R}^n$ onto $\mathbb{R}^n$.*

Proof. For x, y, write $|x| - |y| = D_d(x - y)$, where $d_i = (|x_i| - |y_i|)/(x_i - y_i)$ when $x_i \neq y_i$, and $d_i = 0$ otherwise. Each $|d_i| \leq 1$. Thus

$$H_\sigma(x) - H_\sigma(y) = (C - D_\sigma R D_d)(x - y),$$

which proves injectivity.

Compactness of the regular interval family gives a uniform positive lower bound α on its smallest singular values. Hence $\|H_\sigma(x)\| \geq \alpha\|x\|$. For any b, the continuous squared residual $\|H_\sigma(x) - b\|^2$ is coercive and has a global minimizer x.

Put $r = H_\sigma(x) - b$, and suppose $r \neq 0$. Let J_s range over the finitely many selection matrices whose signs agree with all nonzero coordinates of x. Every convex combination of these matrices lies in the regular interval family. Therefore 0 is not in the compact convex hull of the vectors $J_s^T r$: an equality $0 = \bar{J}^T r$ would contradict nonsingularity of $\bar{J}$. Let $g \neq 0$ be the point in that convex hull nearest to 0. The projection inequality gives $g^T J_s^T r \geq \|g\|^2$ for every adjacent selection. Take $h = -g$. For sufficiently small positive t, the point $x + th$ lies in one such selection region and

$$H_\sigma(x + th) = H_\sigma(x) + tJ_s h, \qquad r^T J_s h \le -\|g\|^2 < 0.$$

The squared residual strictly decreases for sufficiently small t, a contradiction. Thus the minimum residual is zero, proving surjectivity. □

3 Solving the map by a linear program

Fix σ and a rational b. Put

$$P = C - D_\sigma R, \qquad Q = C + D_\sigma R, \qquad X = Q^{-1}, \quad Y = P^{-1}.$$

The matrices P, Q are inverse-M. Thus X, Y have positive diagonal and nonpositive off-diagonal entries. Consider the linear program in the free variable $y \in \mathbb{R}^n$:

$$\min \ \mathbf{1}^T y \quad \text{subject to} \quad Xy \ge 0, \qquad Y(y + b) \ge 0. \tag{1}$$

Lemma 2. *The program* (1) *has an optimum. For every optimum, setting*

$$v = Xy, \qquad u = Y(y + b), \qquad x = u - v$$

gives the unique solution of $H_\sigma(x) = b$.

Proof. Lemma 1 supplies a root x. With $u = x^+$ and $v = x^-$, one has $Pu - Qv = b$. Thus $y = Qv$ is feasible in (1). For every feasible y, $v = Xy \ge 0$ implies $y = Qv \ge 0$. Consequently the objective is bounded below, and its feasible sublevel sets are compact. An optimum exists.

At an optimum, write $u = Y(y + b) \ge 0$, $v = Xy \ge 0$. Suppose $u_k > 0$ and $v_k > 0$. Decrease y_k slightly. The kth entries of u, v decrease by the positive diagonal coefficients Y_{kk}, X_{kk} and remain nonnegative for a sufficiently small decrease. Every other entry weakly increases because the off-diagonal coefficients of X, Y are nonpositive. This gives a feasible strict objective decrease, a contradiction. Hence $u_k v_k = 0$ for every k.

It follows that $x = u - v$ has $|x| = u + v$. Since $Pu - Qv = b$, we obtain $Cx - D_\sigma R|x| = b$. The root is unique by Lemma 1. □

4 The exact hull endpoints

Theorem 3. *For coordinate i, choose $\sigma_i = 1$ and $\sigma_j = -1$ for $j \ne i$. The unique solution of*

$$H_\sigma(x) = b_c + D_\sigma b_r$$

has $x_i = \max_{z \in \Sigma} z_i$. Reversing all these signs gives $x_i = \min_{z \in \Sigma} z_i$. Consequently $2n$ instances of (1) *compute the exact hull.*

Proof. Let $b_\sigma = b_c + D_\sigma b_r$, and let x solve the stated equation. For any true solution z with $A'z = b'$, rowwise interval evaluation gives

$$D_\sigma(H_\sigma(z) - b_\sigma) \le 0.$$

Indeed, a row with $\sigma_j = 1$ uses the minimum possible row product $C_{j,:}z - R_{j,:}|z|$, which is at most b'_j and hence at most the upper right-hand-side endpoint. A row with $\sigma_j = -1$ uses the maximum possible row product, which is at least b'_j and hence at least the lower endpoint.

Write $w = D_\sigma(H_\sigma(z) - b_\sigma) \le 0$. The secant identity from Lemma 1 yields

$$z - x = M^{-1} D_\sigma w, \qquad M \in \boldsymbol{A}.$$

For the upper-endpoint sign choice, row i of $M^{-1}D_\sigma$ is entrywise nonnegative: its diagonal entry is positive, and each off-diagonal entry is a nonpositive entry of M^{-1} multiplied by -1. Thus $z_i \le x_i$. For the reversed sign choice that row is nonpositive, giving $z_i \ge x_i$.

The bounds are attained. Choose any signs s consistent with the nonzero coordinates of x. Then $A^* = C - D_\sigma R D_s \in \boldsymbol{A}$, and

$$A^* x = H_\sigma(x) = b_\sigma \in \boldsymbol{b}.$$

Hence $x \in \Sigma$, proving both endpoint assertions. □

5 Exact complexity and limitations

Each of the $2n$ programs has n variables and $2n$ inequalities. Exact inversion of the rational endpoint matrices P, Q produces polynomial-bit rational coefficients. Polynomial-time rational linear programming [3] and exact postprocessing therefore give polynomial bit complexity for the hull. The interval family is compact and regular, so the solution set is nonempty and bounded for every nonempty interval right-hand side.

The argument does not supply a polynomial algorithm for general regular absolute value equations (AV-03). It uses the nonpositive off-diagonal signs of P^{-1} and Q^{-1} in the coordinate-decrease proof of complementarity. General regularity alone does not provide those signs. The bijectivity lemma alone is an existence result, not a general polynomial-time search algorithm.

References

[1] OpenProblemsInNLA, problem IV-05. `https://github.com/ajt60gaibb/OpenProblemsInNLA/tree/main/intervals-and-absolute-value-equations/IV-05`.

[2] M. Hladík, *An overview of polynomially computable characteristics of special interval matrices*, Springer (2020), 295–310, Section 9. `https://doi.org/10.1007/978-3-030-31041-7_16`.

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A 3×3 interval matrix with at least four real-eigenvalue components

Repository IV-06. Claimed counterexample; independent review pending.

Abstract

A two-parameter interval matrix with integer endpoints has at least four connected components in its real eigenvalue set. Four exact eigenpairs and three exact excluded real numbers provide a short certificate. This disproves the proposed bound of n components for a general $n \times n$ interval matrix.

1 Counterexample

Consider the independent interval entries

$$A(a,b) = \begin{pmatrix} 25 & a & b \\ 1 & -1 & 0 \\ 1 & 0 & 1 \end{pmatrix}, \qquad a \in [-166, -16], \qquad b \in [9, 159].$$

Let

$$\Lambda_{\mathbb{R}} = \{\lambda \in \mathbb{R} : \det(\lambda I - A(a,b)) = 0 \text{ for some admissible } a, b\}.$$

This is the real eigenvalue set in the conjecture recorded in IV-06 [1].

Theorem 1. *The set $\Lambda_{\mathbb{R}}$ has at least four connected components, although the matrix dimension is three.*

Proof. Put $a = -91 + d_1$, $b = 84 + d_2$, where $d_1, d_2 \in [-75, 75]$ independently. Direct expansion gives

$$\begin{aligned} p(\lambda; d_1, d_2) &= \det(\lambda I - A) \\ &= (\lambda - 25)(\lambda^2 + 6) - d_1(\lambda - 1) - d_2(\lambda + 1). \end{aligned}$$

For every fixed real λ, its exact range over the two parameters is

$$(\lambda - 25)(\lambda^2 + 6) + [-75, 75]\big(|\lambda - 1| + |\lambda + 1|\big). \tag{1}$$

There is no interval overestimation: the determinant is affine in each of the two independent parameters and has no mixed term. Thus $\lambda \in \Lambda_{\mathbb{R}}$ exactly when the interval in (1) contains zero.

The following four eigenpairs certify four included real numbers. Each vector is nonzero, each parameter pair is in the interval, and direct integer multiplication verifies $A(a,b)v = \lambda v$.

λ	a	b	v^T		
-3	-21	154	-4	2	1
0	-16	9	-1	-1	1
3	-146	29	4	1	2
25	-91	84	312	12	13

At three intervening real numbers, (1) yields

λ	exact determinant interval
-1	$[-332, -32]$
1	$[-318, -18]$
12	$[-3750, -150]$.

None contains zero. Therefore

$$-3 < \underbrace{-1}_{\notin \Lambda_{\mathbb{R}}} < 0 < \underbrace{1}_{\notin \Lambda_{\mathbb{R}}} < 3 < \underbrace{12}_{\notin \Lambda_{\mathbb{R}}} < 25$$

places the four included points in four distinct connected components: every connected subset of the real line containing two points must contain all points between them. This proves the theorem and disproves the proposed general bound. □

2 Scope and reproducibility

Only two matrix entries vary, and their uncertainty is independent. All other entries are fixed integers. The argument proves at least four components; locating every component endpoint is unnecessary for the counterexample. It concerns the union of real eigenvalues of all admissible real matrices, not the complex spectral set and not an enclosure returned by an interval algorithm.

The accompanying exact test script expands the characteristic polynomial, verifies all four integer eigenpairs, and computes the three determinant intervals using rational arithmetic. No floating-point eigenvalues are used in this certificate.

References

[1] OpenProblemsInNLA, problem IV-06, including its original conjecture references. `https://github.com/ajt60gaibb/OpenProblemsInNLA/tree/main/intervals-and-absolute-value-equations/IV-06`.